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Participatory Planning and Gamification: Insights from Hungary

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Abstract: Involving citizens in public decision making has become more and more important recently. However, activating citizens is challenging in urban planning, especially in post-socialist countries, such as Hungary, where civil activity is weak, and citizens' attitudes are characterized by distrust towards decision-makers. The gamification of planning processes aims to address this issue and support a more democratic planning process. Gamification is the application of game-like elements (e.g., rewards, storytelling, feedback, competition, etc.) in non-game contexts, thus transforming them into more engaging and enjoyable activities. This study aims to present how gamification is used in the Hungarian urban planning processes and the obstacles to using gamified practices. The research is based on semi-structured interviews with urban planning experts in major Hungarian cities. Our results show that although decision-makers are open to using participatory approaches and gamification techniques, no clear guidelines and principles support these aspirations. The lack of trust and the apathy of citizens makes participatory planning a challenging task. Furthermore, the lack of resources is an obstacle and motivation at the same time when innovative planning methods are used.

Keywords: gamification; urban planning; Hungary; participation; citizen engagement; post-socialism; interviews

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1. Introduction

Gamification strategically integrates game design elements in non-game contexts to boost motivation and engagement [1]. In recent years, particularly over the last decade, gamification has gained significant traction in participatory planning. Notably, studies by Gordon and Baldwin-Philippi [2], Aguiar-Castillo et al. [3], Thiel and Lehner [4], and Mandujano et al. [5] highlight the growing importance and effectiveness of these techniques in fostering community involvement and enthusiasm in urban planning projects.

The adoption of gamification in urban planning reflects a shift in approach, moving away from traditional methods of public engagement to more innovative and interactive practices. By incorporating elements such as point scoring, competition, and rewards, planners are finding new ways to capture the interest of citizens, encouraging them to participate actively in the decision-making processes that shape their communities [6,7]. This evolution in engagement strategies signifies a broader recognition of the value of citizen input and collaboration, ultimately contributing to more inclusive and responsive urban environments.

Integrating game elements into presenting serious societal issues can considerably enhance citizen motivation and engagement. By transforming complex topics into more

accessible and interactive experiences, game-based approaches can facilitate greater participation among community members [8]. This is particularly important in the context of participatory urban planning, where the persistent problem of citizen demotivation drives the need for innovation in methodologies. Addressing this challenge is essential for fostering meaningful involvement and ensuring that urban development reflects the needs and aspirations of the community. Furthermore, gamification is also a learning process for all the parties involved. Thus, the effects of such initiatives go beyond opinion polls or contributions to the planning and implementing of particular programs or projects [9].

The pressing challenge we encounter on a global scale is the widespread belief among citizens that their ideas and requests are often ignored or undervalued by local governments. This pervasive sentiment not only diminishes civic engagement but contributes to a substantial erosion of trust in public institutions. Many individuals arrive at this conclusion based on their negative experiences with local government interactions, where their voices may have gone unheard or unacknowledged. As a result, this has created a significant disconnect between residents, governmental bodies, and various stakeholders. This gap poses profound implications for community collaboration, effective governance, and the overall health of democratic processes [10].

In recent decades, Hungary's urban planning field has undergone substantial transformation, reflecting evolving societal needs and priorities. This transformation has been marked by a progressive shift towards participatory approaches, prioritizing citizen engagement in urban planning. These methodologies aim to empower individuals and communities by inviting their active participation in decision making, ensuring that diverse perspectives and local knowledge are integrated into planning outcomes. This trend not only fosters greater transparency and accountability in urban development but also enhances the overall effectiveness of urban policies by aligning them more closely with the needs and aspirations of the population [11,12]. Nonetheless, the previously mentioned gap continues to be challenging in this context. However, incorporating gamification as an innovative approach in participatory practices could bring the actors closer together and create a connection between them.

This paper aims to deliver an overview of the current landscape of urban planning in Hungary, focusing specifically on the participatory practices employed within this context. The paper seeks answers to three questions. Our first question was as follows: what is the current state of participatory urban planning in Hungary? The second question was the following: how is gamification used in Hungarian participatory urban planning? Finally, our third question is, what are the obstacles to gamified urban planning in Hungary? To answer these questions, we conducted in-depth interviews with professionals in the field and government officials involved in urban planning processes. The goal was to explore how these participatory approaches shape urban development, informing future policymaking and providing insights into the effectiveness and challenges of public participation in the Hungarian urban planning framework. The paper is structured as follows: first, we review the literature on participatory planning and gamification. In the following sub-section, we highlight the key features of the Hungarian context. The next section describes the process of data collection. Then, we present our findings within the last two sections, and we interpret our results in the light of the body of previous research and highlight the theoretical and practical implications (Figure 1).

To gain a comprehensive understanding of contemporary urban planning trends through participatory methods, it is essential to provide a succinct theoretical background that includes general attributes and those specific to Hungary.

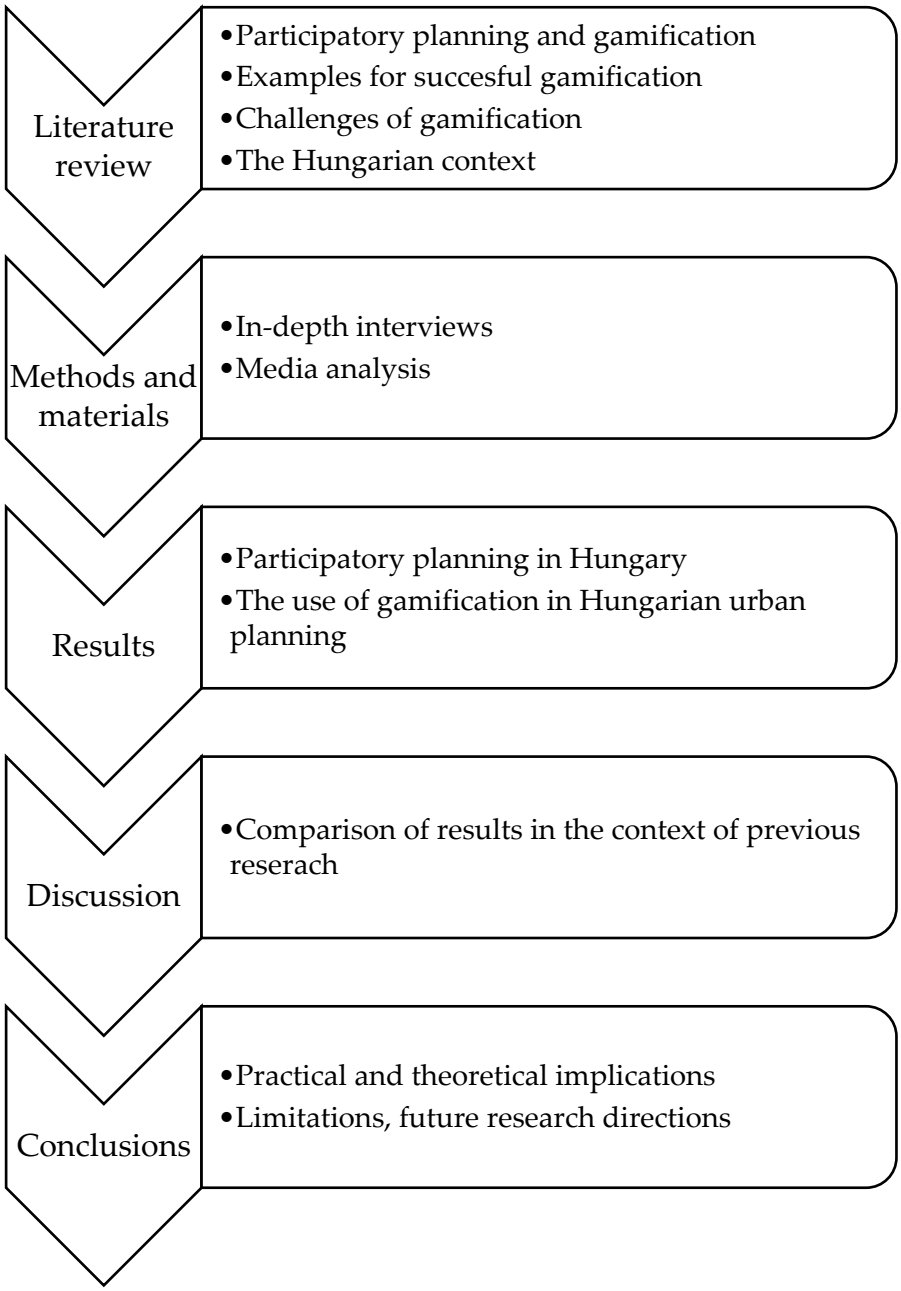


Figure 1. The structure of the paper. Elaborated by the authors.

2. Literature Review

Urban planning has increasingly been scrutinized for its propensity to perpetuate the existing inequalities and adhere to the established trajectories. Considering this criticism, there has been a growing emphasis on the necessity for planning processes that are more responsive, adaptable, and participatory. This evolution has resulted in various theoretical frameworks designed to address these challenges, including collaborative planning, transition management, co-creation, and design thinking [13].

Collaborative planning, influenced by Habermasian concepts of ideal speech, emphasizes the importance of stakeholder engagement and decision making in complex situations. Stakeholders with varying views, power dynamics, and levels of knowledge need to participate in deliberative processes to reach a consensus and minimize conflicts, as illustrated by the "Stuttgart 21" project case where the lack of communication and transparency and the

lack of deliberation between stakeholders led to financial damage and delay to the project, along with protests [7].

However, an alternative perspective in the literature argues that stakeholder engagement should not be seen merely as a moral obligation but as a key component of a firm's strategic objectives [14]. Freeman's stakeholder analysis suggests that strategic decision making should consider the interests of a wide range of stakeholders, not just those with direct economic influence. Including "powerless" stakeholders in the analysis phase can provide valuable insights into the values and priorities that ultimately impact the final decision [15]. Furthermore, secondary stakeholders who advocate for primary stakeholders may play a vital role in synthesizing diverse viewpoints.

While the transparency and inclusivity of collaborative planning are essential factors, the strategic significance of stakeholder engagement must also be considered. This ensures that decision-making processes remain aligned with the organization's broader objectives rather than being overly narrow. Despite its merits, collaborative planning has been criticized for its focus on consensus-building, which may not always be practical or desirable in addressing deeply contested urban issues. A balanced approach that integrates stakeholder engagement's moral and strategic aspects may lead to more effective and sustainable urban planning outcomes.

Transitions management, on the other hand, takes a complexity-informed, normative approach to sustainable futures, emphasizing experimentation and adaptive pathways. This approach recognizes the inherent complexity of urban systems and the need for flexible, iterative planning processes. Design thinking, derived from the design disciplines, strongly emphasizes stakeholder engagement and collaborative decision making. Unlike collaborative planning, design thinking encourages a more experimental and iterative approach, focusing on prototyping and testing solutions [13].

While these three approaches share a common goal of fostering more participatory and transformative planning processes, they differ significantly in their theoretical foundations, methodologies, and priorities [13,16].

The increasing popularity of these participatory approaches, particularly in urban sustainability transitions, has led to a growing interest in gamified participatory planning [17]. The integration of gamification in urban planning has become an increasingly prominent focus in recent years. This trend is driven by the desire of municipalities and urban development professionals to find innovative strategies for engaging citizens in the planning process. By incorporating game-like elements, such as point scoring, competition, and interactive platforms, urban planners aim to enhance community involvement and ensure that residents have a voice in decisions that impact their neighborhoods [8,18,19]. Furthermore, gamification can improve decision making by simulating various urban development scenarios, allowing citizens to visualize the potential outcomes of different planning choices. This fosters a deeper understanding of urban issues and encourages collaboration among community members. Ultimately, the goal is to promote more sustainable and livable communities by aligning development initiatives with the needs and preferences of the residents they serve [18,20,21]. Various studies highlighted that gamification as a democratic innovation can serve as an empowering tool for citizens [22,23]. Gamification enables citizens to imagine and experiment with social innovations, creating and testing ideas for a more sustainable or democratic future [24].

The global integration of gamification in urban planning has gained momentum due to a growing recognition of its significant potential to enhance public engagement and participation. By incorporating game-like elements into planning processes, municipalities can foster an environment that encourages community involvement, allowing residents to actively contribute their insights and preferences [19,23–26].

Additionally, gamification serves to improve data collection by transforming complex information into more accessible and engaging formats, thereby encouraging a wider range of participants to provide valuable input. This innovative approach not only aids in gathering diverse perspectives but also improves the quality of data collected [21,27,28]. However, the flow of information is not one-directional: there is a mutual learning process and the co-creation of new knowledge [29]. Also, design alternatives can be tested using gamification, as Ehab and Heath [30] demonstrate through the example of urban design projects in London, where virtual reality was used as a co-design tool.

Furthermore, gamification facilitates collaborative decision making by creating interactive platforms that enable stakeholders, including citizens, planners, and policymakers, to work together more effectively. Urban planning can become more inclusive and adaptive through these collaborative efforts, leading to more sustainable and successful outcomes [31,32].

2.1. Successful Gamification

Various case studies that have employed gamified methodologies to enhance citizen engagement highlight the increasing importance of gamification in urban planning [33]. For instance, Kazhamiakin et al. [34] investigated using a service-based gamification framework in Rovereto, Italy, to encourage sustainable urban mobility. The study emphasizes gamification's potential to promote voluntary behavioral changes among citizens, a vital aspect of smart city initiatives.

Similarly, Gabrielli et al. [35] describe the design and implementation of gamified systems in several European cities, including Trento, Milan, Helsinki, and Barcelona, to involve citizens in sharing information about parking spaces and to encourage sustainable commuting habits. Kazhamiakin et al. [36] present a report on a successful gamified system in Eindhoven, Netherlands, which was developed to motivate employees to choose sustainable commuting options. Another noteworthy example is creating a gamified mobile application called WasteApp as a part of an EU-funded project. This application aims to promote recycling behavior among citizens by incorporating game design elements, and the study evaluates various factors influencing the intention to use this gamified solution [3].

In Sweden, the innovative game Cities: Skylines was harnessed through dynamic workshops to aid in planning the vibrant central area of Norra Djurgårdstaden. This engaging tool transformed the urban planning process into an interactive experience, allowing residents to immerse themselves in the creative design of their community. Through this collaborative platform, individuals were empowered to articulate their visions and suggestions, shaping their neighborhood's future in a meaningful and impactful way [37].

While the potential benefits of gamification in urban planning are clear, it is also essential to consider critical perspectives. A viewpoint paper titled "Cities and the Politics of Gamification" underscores the need for a deeper understanding of the cultural and power dynamics involved in the gamification processes within urban contexts [38].

2.2. Challenges of Gamifying

However, the application of gamification in urban planning does come with challenges. Souza et al. [39] offer a critical perspective on the political and cultural implications of gamification in cities, warning against the risk of "gamipulation"—where game design elements can be used to influence user behavior. Nonetheless, the overall evidence from the case studies suggests that gamified urban planning practices can serve as a valuable tool for fostering citizen engagement, promoting sustainable behaviors, and enhancing the livability of cities. At the same time, gamification can also have adverse effects: it can compromise the quality of citizen participation or enhance competition instead of collaboration. In addition,

gamified techniques are more likely to engage certain demographic groups (mostly younger generations) [33].

Cultural contexts play a crucial role in adapting new techniques and approaches to urban planning. In the post-socialist region, civil activity was weak as a legacy of the totalitarian era. It resulted in low trust in public institutions, weak community engagement, and a lack of social cohesion. As a result, a certain "participatory apathy" was a widespread feature of these societies [40–42]. However, the urban planning landscape in Hungary has undergone significant transformation in recent decades, characterized by an increasing emphasis on participatory approaches that seek to engage citizens actively in the decision-making process [11,12,43,44]. This shift has emerged from the recognition that traditional top-down planning models frequently fail to address local communities' diverse needs and perspectives adequately. Consequently, a more inclusive and collaborative methodology is essential for creating sustainable and livable urban environments. Furthermore, participation became crucial due to the lack of financial resources at the local level. The centralized political system decreased local authorities' revenue; thus, mobilizing non-financial resources through increasing citizen engagement is crucial.

2.3. The Potential of Applying Gamification in Hungary

The transition toward enhanced civic engagement in urban development has been an evolutionary process, reflecting a broader societal movement towards democratic and participatory governance models in post-communist Hungary. This transformation has been influenced by several factors, including the decentralization of political power after the regime change, the growth of civil society organizations, and an increasing acknowledgment of the significance of public participation in shaping urban spaces [45]. Various initiatives and mechanisms have been established to promote greater citizen involvement in the planning and decision-making processes. These efforts aim to ensure that the needs and aspirations of local communities are more effectively represented in the development of their urban environments [46,47].

The evolution of participatory planning practices in Hungary can be linked to the post-communist transition. This transformative period commenced in the late 20th century as the country embarked on the journey towards a more democratic framework. This transition was characterized by significant sociopolitical changes, which prompted a reevaluation of government structures and public policy approaches [11]. During this crucial time, there was an increasing recognition among policymakers and urban planners of the essential role that civic engagement plays in shaping urban environments. Stakeholders began to understand that involving citizens in the planning process enhances the legitimacy of decisions and ensures that developmental plans reflect the needs and desires of the community [48,49].

In response to this growing awareness, various initiatives and programs were launched to foster greater public participation in urban planning and decision making. These efforts included community workshops, public forums, and collaborative planning sessions, all designed to encourage dialog and input from citizens. Such initiatives aimed to create more inclusive and transparent governance structures, ultimately contributing to Hungary's cities' and communities' long-term sustainability and vitality. This shift towards participatory planning has played a pivotal role in shaping contemporary urban development in the country [50,51]. However, practitioners have often experienced citizen mobilization not being easy due to cultural features and the lack of know-how (best practices, skills, approaches, etc.) [52]. As the examples presented above, gamification can overcome these challenges.

Gamification has emerged as a practical approach to bolstering participatory planning practices in Hungary. By integrating play, competition, and reward components, gamification can make the planning process more engaging and accessible to a broader spectrum of stakeholders, including those historically marginalized from such processes [53,54]. Gamification can increase citizen engagement and involvement in urban planning by making the process more interactive, fun, and rewarding. It can incentivize participation, encourage collaboration, enhance the flow of information between various actors, and foster a sense of community ownership over the development of cities. Using game-like elements such as points, leaderboards, and badges, gamification can motivate citizens to contribute ideas, provide feedback, and actively participate in the planning and decision-making processes that shape their local environment [55–58].

After 2010, the Hungarian government's recentralization processes transformed the urban planning environment. This process has meant centralizing financial resources and forcing municipalities to be more effective and use as many resources as possible [59]. Hungary's changing urban planning landscape suggests that incorporating gamification and other participatory methods will be essential for shaping the future of the country's cities. Implementing these approaches can create a more inclusive and collaborative planning process. This empowers citizens to engage in the development of their communities actively, ensuring that the resulting plans better represent the needs and aspirations of residents [54]. At the same time, it is worth highlighting that the authoritarian tendencies within the political sphere have strengthened Euroskeptic attitudes and generated a certain level of mistrust against "external" ideas—such as gamification.

The literature highlights two main characteristics: first, participatory methods have recently gained attention internationally and in Hungary; second, while gamified methods are not yet widely used, their effectiveness has been demonstrated. Based on the interviews conducted with professionals, these findings provided the foundation for our research.

3. Materials and Methods

First, we identified the scope of this study, which is the presence of gamification in participatory urban planning practices. After that, we defined the national context for our paper. In Hungary, this is a little-researched topic, which made it necessary to deal with it. Above all, the Central European context of the study could contribute to the research of the neighboring countries as well. According to the literature, participatory methods and gamification can be studied through various methods. The experiences of locals are often analyzed through questionnaire surveys, while experts' viewpoints can be revealed both through surveys and interviews (e.g., [18,60]). Case study analyses (e.g., presenting the resources used or analyzing the results of the actions) are also used to reveal the effects of gamification (e.g., [5,48]). Studies also analyzed the available or work-in-progress applications to present the tools of civic engagement (e.g., [3,22]). Testing games and analyzing recorded game sessions were also used to understand how gamification affects decision making (e.g., [19,25,61]). Since our study is exploratory research and aims to analyze the general tendencies of participatory planning and gamification in Hungary, acquiring information from experts was the most important task.

Our initial attempt was to conduct an online questionnaire; however, it was unsuccessful. The failure occurred due to insignificant responses preventing us from concluding. Consequently, we have determined that in-depth interviews are needed instead to uncover the current state of participatory urban planning and its relation to gamification in Hungary. According to Tuza and Kovács [60], this method enables a deeper and more nuanced understanding of the processes, revealing the perceptions, attitudes, and opinions of actors. In addition, we also conducted focused media research to identify the discourses

in relation to participatory budgeting, which is one of the most often used participatory tools in Hungarian planning practice. The aim of the media analysis was to supplement the information acquired through in-depth interviews.

For our research, we conducted semi-structured interviews with government officials and professionals from organizations implementing participatory and/or gamified methods in urban planning. Interviews provide opportunities to reveal the experiences of professionals not just regarding their work but also the broader processes (e.g., the general status of gamified urban planning in Hungary, the results, and challenges arising in other municipalities). We selected interviewees who are professionals based on the criteria that they have work experience in participation and are currently employed in their field. After choosing the first interviewee, their insight led us to use the snowball sampling technique and contact the following partners (Figure 2).

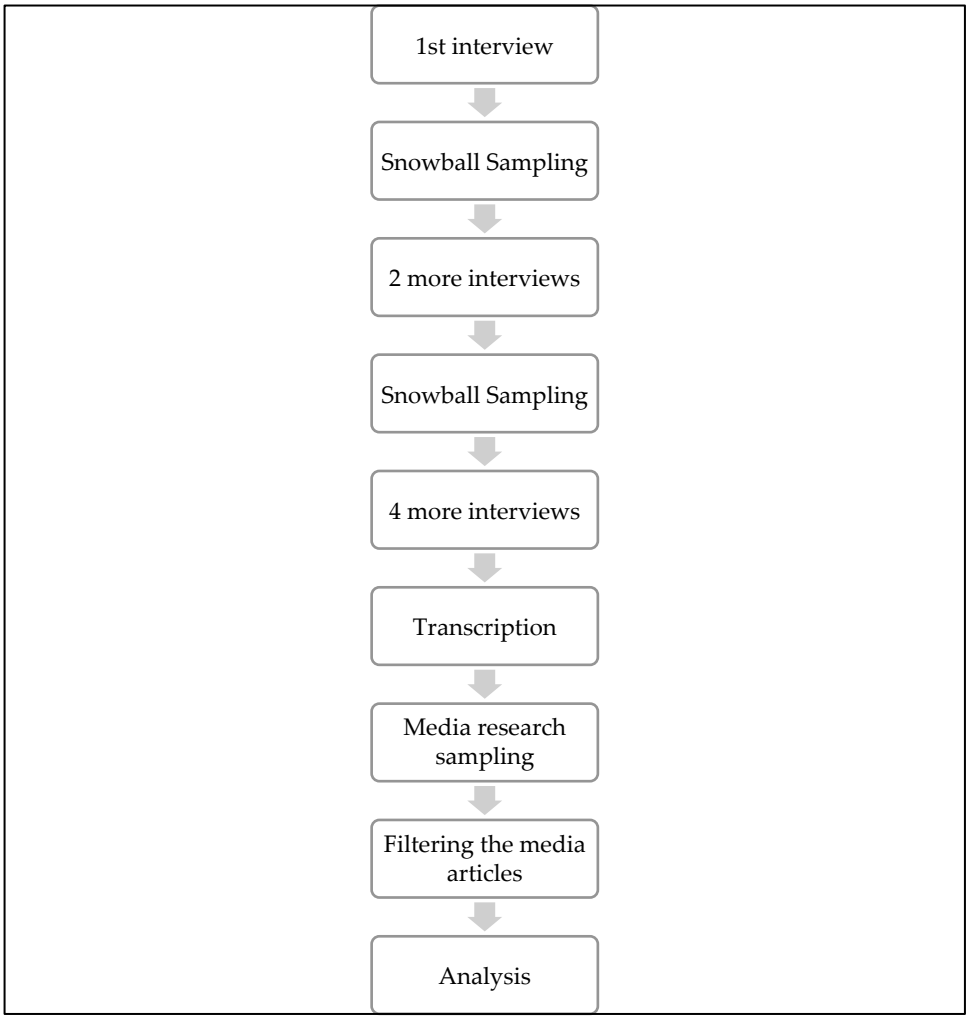


Figure 2. Process of interviewing. Edited by the authors.

These interviews were tailored to each interviewee and focused on three main areas: their work experience related to participation, their understanding of gamification and gamified methods, and their likelihood of using gamified approaches. Additionally, the discussions explored the potential usefulness of a collection of gamification best practices. In most cases, the discussions raised additional questions not covered in the guide, allowing for a more thorough exploration of the topic. The interviews were recorded and transcribed; then, these transcriptions were analyzed. Following the analysis, we formulated the study's theoretical and practical implications (Figure 3).

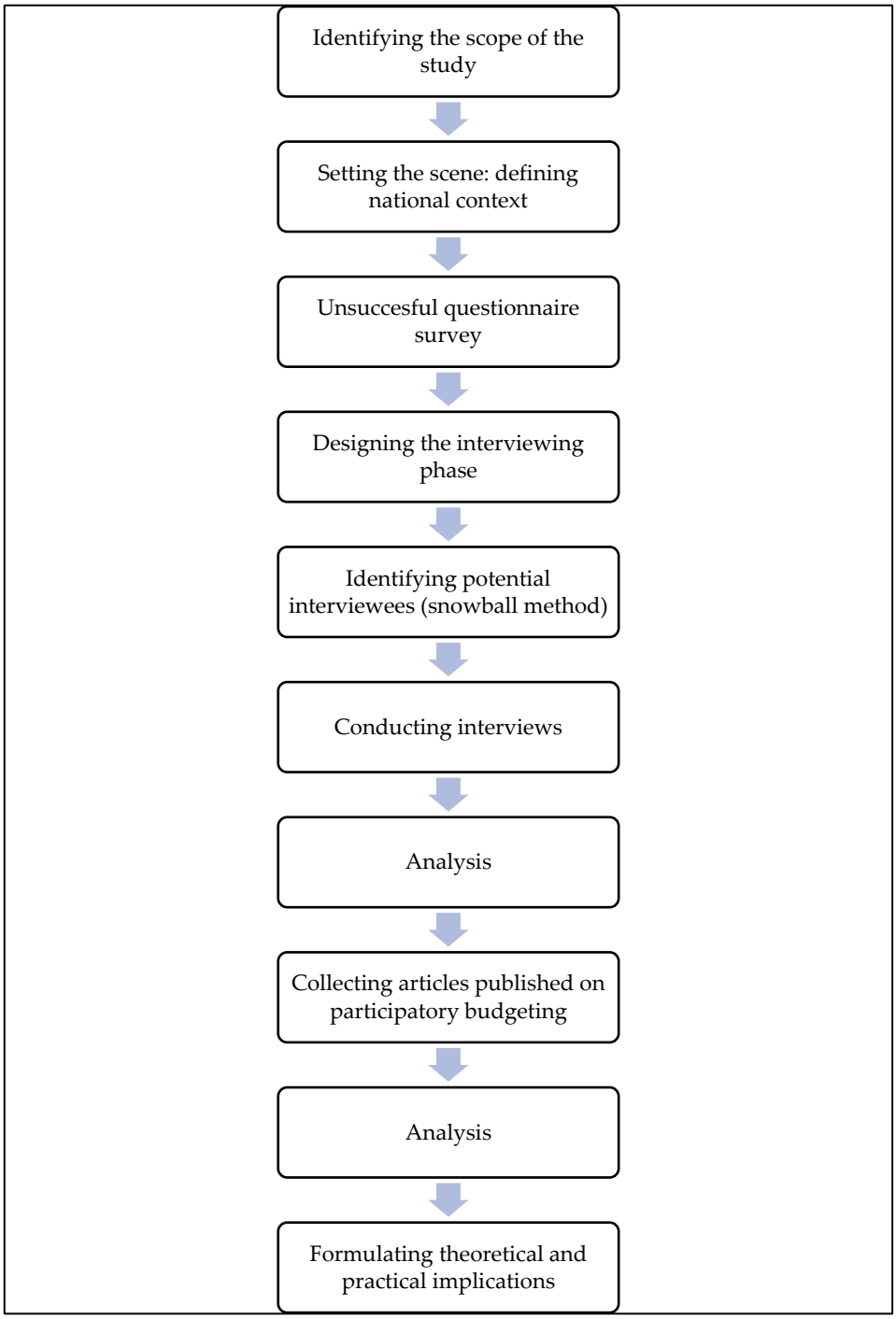


Figure 3. The course of the research. Edited by Lajos Boros.

During the study period, from September 2024 to December 2024, we conducted seven interviews with an average length of 35 min, where the minimum time was 25 min, and the maximum time was 45 min. The interviewees were informed about the research, and we assured them of complete anonymity.

Although it is uncommon for such a small sample size to be significant, it is essential to note that both gamification and public participation are relatively recent phenomena in Hungarian urban planning practices. As a result, there is a limited number of potential interviewees. However, the interviewees possessed valuable insights into the functioning of municipalities at various settlement levels (Table 1). Several analyses deal with the issues of small interview samples, highlighting the usability of research with few participants. We

used "purposive sampling" [62], selecting interview partners with extensive knowledge of their city and the whole Hungarian planning scene. Thus, they also provided general information beyond describing the processes within their municipalities. According to the analyses (e.g., [63–65]) focusing on the appropriate sample sizes in interview research, the scope and the approach of the study play an important role. Our research aims to uncover the general trends and experiences regarding participatory planning and the role of gamification in it. This scope requires experts on the field, whose number is limited—in a sense, the sampling has a resource constraint [66]. On the other hand, some studies point out that in many interview research, the first interviews generate most of the data [67]. Thus, Guest et al. [68] state that 6–12 interviews can be enough for research in the case of a homogenous sample. Our group of interview partners was homogenous because they all represented the professional planning side of the processes. Thus, although the sample sizes can consistently be increased, our sample still provides insights into the analyzed topic.

Table 1. Interviewees by occupation and workplace. Edited by Katinka Tóbiás.

Interviewee	Profession	Workplace
Interviewee No. 1.	project lead	non-profit organization
Interviewee No. 2.	municipality officer	local government, major city
Interviewee No. 3.	municipal representative	local government, major city
Interviewee No. 4.	community organizer	local government, major city
Interviewee No. 5.	program leader	climate policy institute
Interviewee No. 6.	municipal representative	local government, major city
Interviewee No. 7.	consultant	non-profit organization

Our main challenges during the interviews included reaching out to individuals and coordinating suitable discussion times. Additionally, since most interviews were online, connectivity issues occasionally disrupted conversations.

To supplement the results of the interviews, media research was also conducted (Figure 4). We collected articles that focus on participatory budgeting published in online media outlets. We focused on this topic because, according to the interviews, this is the most well-known and most often used participatory method among Hungarian municipalities. The following keywords were used (in Hungarian): participatory budget interview, participatory budget evaluation, and participatory budget opinion. Our search returned 412 results. Since our aim was to identify the related discourses, we only analyzed those articles that presented interviews with decision-makers. Thus, articles describing the method or the background of participatory budgeting, presenting the calls for participation, or describing the submitted ideas or results of the voting were left out of the analysis. Altogether, 31 articles remained in the sample after the filtering. The low number of articles after filtering demonstrates the novelty of participatory methods and highlights the lack of knowledge on the topic within society. The articles made it possible to gain more information on the attitudes of decision-makers and to supplement the results of the in-depth interviews.



Figure 4. The process of media analysis. Elaborated by the authors.

We conducted a qualitative analysis of the articles focusing on the following topics: How are participation and participatory budgets presented? What are the expected benefits of participation? What kinds of results and critiques are mentioned?

4. Results

The main results of the interviews are presented in this chapter, focusing on the attitudes and experiences towards gamification and participatory planning. According to the literature, participatory methods are becoming increasingly common in urban planning, and Hungary is no exception. Our results support this statement since all the interviewees were familiar with participation and had previously attended participatory urban planning events. They unanimously agreed that these methods are becoming more mainstream:

> "*. . .participation is becoming very fashionable recently, and we like to use this word participation, and we can also insist on every process that it is participative. . .*" (community organizer, major city)

According to the media analysis, one of the key reasons behind this process is that decision-makers aim to bridge the widening gap between policymaking and local societies:

> "*Everywhere in the world we experience the professionalization of policymaking. This separates locals, decision makers and politicians. That is why citizen assemblies are initiated [. . .] to discuss issues in various topics and formulate policy proposals. Participatory budgeting is about helping the creation and implementation of ideas that are invisible for officials but contribute to the well-being of residents.*" [69]

However, some have argued that this process is much slower than it appears (and should be):

> "*Based on our experience, it is starting to develop very slowly in Hungary, so that local governments start using it.*" (program leader, climate policy institute)

The contrasting perceptions of participation in Hungary stem from the diverse viewpoints held by various stakeholders involved in urban planning. Each actor, whether they are government officials, community members, or urban planners, approaches the idea of participation from unique angles, influenced by their experiences, interests, and objectives. This variance shapes the overall dynamics of how urban development initiatives are conceived and implemented, highlighting the complexity of collaboration in the urban environment.

Another intriguing finding from the interviews was a prevalent skepticism regarding the government and its stance on various issues. This skepticism starkly contrasts the growing adoption of participatory methods, which suggest a push for more inclusive and engaged governance. The tension between the public's distrust and the idea of collaborative decision making highlights a complex relationship between citizens and their government:

> "*. . .basically, a municipality has long-term visions, and it's good if it involves the local population, so this in itself is point zero, that we don't really expect municipalities to want or be able to fulfill this role.*" (community organizer, major city)

This issue is closely related to the broader challenge of balancing the interests of the public with those of local government authorities. Finding common ground is crucial, as it involves addressing the concerns and needs of the community while ensuring that government policies and decisions align with their objectives:

> "*Therefore, we always say that one of the most difficult tasks of working in local government, but also the most beautiful challenge, is to somehow reconcile the very diverse needs and opinions of the population and, with that, to find compromises that obviously represent the majority opinion*." (municipal representative, major city)

This challenge may stem from or intersect with a common communication issue and a lack of transparency. The interviewees all agree that communication is essential, yet it is a long process of building up the public's trust:

> "*. . .actually, the operation of the local government should be made participatory, which has a lot of not so spectacular parts [. . .] so what is not that you can come to the general assembly or that its materials are public, but that the local government actively informs you about what is happening.*" (municipality officer, major city)

> "*and let's say the people who come to the forum experience a very honest communication, where we don't use such political election texts, but we really tell you what the situation is and what could be done and what the difficulties are and what we want, but it may not be possible to do it and then facilitated in a workshop-like manner, they can actually say what their needs are.*" (municipality officer, major city)

On the other hand, there are some cases where the local government tries to apply all possible forms of communication to reach more people:

> "*. . .we appealed to the population through all kinds of channels: the municipality's website, Facebook page, representatives' Facebook pages, local press, print media, TV.*" (municipal representative, major city)

> "*We use a lot of different channels, newsletters, website, Facebook, Instagram, and then the offline posters, there are biger info points with small signs, and mini-info points that are posted on the fences of all such public areas.*" (community organizer, major city)

However, they often struggle to find effective communication methods and the necessary human and financial resources:

> "*So that we have very few communication channels. Actually, mostly Facebook, which is a joke somewhere.*"— (municipality officer, major city)

> "*They don't always have an effective communication channel with the population, but even if they did, the other big obstacle is obviously money. So that they are usually forced to the minimum required.*" (program leader, climate policy institute)

The lack of resources endangers the sustainability of participatory methods, for example, in Budapest or one of the major cities, Győr, where due to the austerity policies of the national government, the local authorities face difficulties in providing the required funds for the participatory budget.

> "*. . .It is a difficult situation, since due to the government cutbacks, the participatory budget can be initiated only as a pilot project with a much smaller budget.*" [70]

Having limited funds not only decreases the magnitude of the projects but sustains a low level of participation. According to some experiences, the more money is spent through participatory budgets, the higher the participation rate among locals [71]. Some have recognized that the effectiveness of participative initiatives largely depends on their scale. Engaging smaller communities, particularly at the neighborhood level, is generally more straightforward, as this requires fewer financial and human resources. In many cases, the focus of these initiatives revolves around specific aspects of the community, such as the renovation of public spaces, which directly impacts residents (Figure 5).

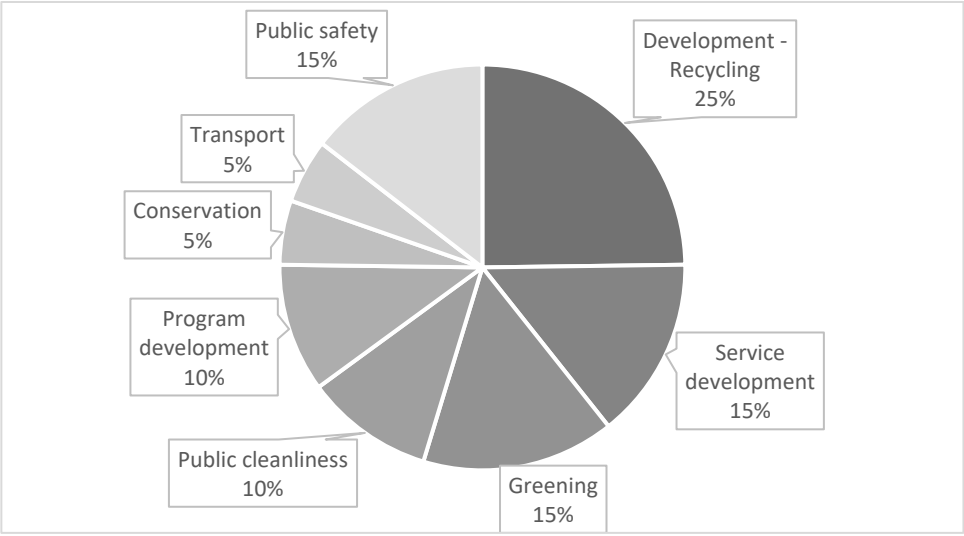


Figure 5. Distribution of votes of residents cast by idea groups. Edited by Katinka Tóbiás based on the document of the VIII. District of Budapest called "Numbers and the Józsefváros participation budget." Source: https://jozsefvaros.hu/downloads/2022/12/rk_adatbazisok_elemzese-pdf.pdf?ver=20221220074612, accessed on 27 February 2025.

Consequently, it is more likely that residents will participate when a development affects their area.

> "*But you also must see that these are primarily local things. So, the local community is the one really interested in the renovation of a public park or a playground.*" (municipal representative, major city)

This also makes it easier to reach out to the public in different ways, such as the info signs or points mentioned above, as well as by sending letters or personal visits.

Based on the experiences of municipal representatives from major cities and given the scale at which they think, smaller settlements or towns may not require participative or gamified methods for community engagement. This is likely because the residents know each other well, and the mayor is familiar with the community. There may be little need to motivate people to communicate in small villages and towns, as they tend to engage naturally. However, this conclusion is merely an assumption based on the interviews and would require further research for validation.

Another key aspect of the interviews was motivation, specifically the methods used to encourage public participation in urban planning. This was highly important considering gamification—the main topic of the study—since it serves as a tool to increase motivation. When we analyzed the qualitative data, we found different ways the interviewees used or knew as a motivational tool that led to other perspectives on the matter:

> "*. . .perhaps this is the best way to encourage residents to give them the opportunity to express their opinions on as many channels as possible.*" (municipal representative, major city)

> "*Perhaps the other incentive, which is again not a classic incentive, is to present the renovation processes where fellow residents participated in their opinions so that they can see that, yes, what your neighbor said has come true [. . .] so perhaps we can also motivate residents to participate, if you see good examples of how other opinions have been given and the municipality has accepted them, then there is no danger, as you mentioned, that it will not happen here anyway, and why should I participate in this matter*"— (municipal representative, major city)

> *"Well, if we think about the project, that Foodwave, the food waste event, someone made the video and won an iPhone and an iPad. We took the whole team to Vienna on a study trip."* (project lead, non-profit organization)

The media analysis supported these findings; for example, the most often emphasized aims of participatory budgets were to encourage public participation and to strengthen local democracy.

> *"We wanted to bring decision-making closer to the locals, thus they can be more familiar with the processes. This changes their perspective on the work done by the municipality, and finally it also increases their trust between decision-makers and locals."* [72]

> *"We often describe the voting on the ideas as democracy project—since it is not enough to submit an idea, gaining support is also crucial. Collaboration is required to have the support of neighbors."* [73]

The interviews indicate that idea submissions, competitions (i.e., the work of Budapest Tuning), and workshops are standard. While there are larger gamified projects, they are not clearly labeled as such and seem to appear unknowingly. Some examples of that are from the capital city Budapest, such as the Budapest13 Smart City mobile app, which helps connect the citizens of XIII. district with the municipality through mobile parking, mobile tickets, district news, and prize games (Figure 6).

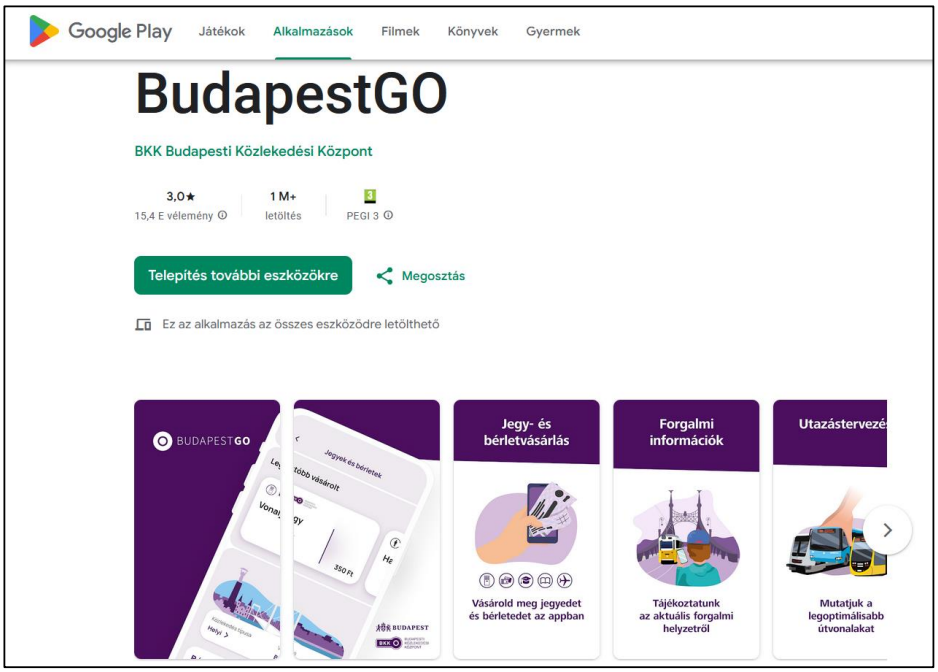


Figure 6. Budapest13 Smart City application on GooglePlay. Source: https://play.google.com/store/apps/details?id=hu.webvalto.bkkfutar, accessed on 7 March 2025.

Another notable example is the Foodwave Óbuda initiative, where participants were required to upload pictures or videos related to food waste and its reduction. As highlighted in the quote, participants could win prizes and go on a field trip.

In another instance, a drawing competition for children was held, with the winner's artwork being displayed on the mosaic of a newly renovated public square.

Another notable instance is the "Green Hackathon" (Figure 7). This event aimed to encourage teams to compete in creating the most sustainable and green ideas for their city. The winner received a prize, and their ideas were incorporated into the final conception.



Figure 7. Moment of the Green Hackathon. Source: https://www.delmagyar.hu/helyi-kozelet/2024/06/a-varos-zolditesevel-kapcsolatosan-rendeztek-otletversenyt-galeria, accessed on 29 January 2025.

Only one individual was initially unfamiliar with gamification but became open to it once they learned more about it. They recognized the potential in it regarding citizen mobilization:

> "*I think it is likely that this would increase the number of participants.*"— (municipal representative, major city)

As the media analysis revealed, participatory actions are sometimes considered political actions, and their main aim is to increase the popularity of the local mayor. Other critics highlight that the implementation of projects is often delayed, and participation is quite low.

> "*Only 1.7% of the locals participated in the formulation of participatory planning in the 8th District of Budapest. There is a general disinterest towards such projects, thus they are just wasting the money.*" [74]

Some argue that the ideas raised through participatory methods do not bring innovation to urban planning either:

> "*While the method itself is innovative, most of the ideas are quite traditional. The most supported proposals are related to children, dogs, bicycles or trees. In some cases, locals voted for creating sidewalks or pedestrian crossings.*" [75]

On the other hand, the advocates of participatory methods emphasize that the methods require time: since locals are not accustomed to having real opportunities for participation, it is difficult to mobilize them.

> "*This is a slow democracy tool that is useful to change the apathy of people and their subordinate role in public matters*" [76]

There is some skepticism regarding the potential use of gamification to enhance motivation as well. Our observation suggests that this hesitance stems from a lack of familiarity with gamification in the context of urban planning, as most examples they have encountered are primarily focused on education. This limited exposure may contribute to their reluctance to explore gamification as a viable tool in urban planning initiatives. In summary, while they know gamification and may attempt to incorporate some games or playful elements into their events or citizen engagement efforts, they do not know how to implement it effectively.

Thus, the consensus among the participants was that they were likely to use gamification; however, a significant concern was the absence of established guidelines to help them

apply this new method effectively. As a result, all the interviewees agreed that compiling adaptable best practices on gamification would be instrumental.

> "*. . .the one thing that is useful is if there is such a toolbox, and actually, it might not even be a toolbox, but somehow, so this gamification should obviously have such a definition background. . .*" —(project lead, non-profit organization)

To overcome the difficulties concerning the lack of know-how, civil organizations can play a significant role in helping local governments implement gamification. These organizations often use more gamified approaches and have valuable experience that can serve as an excellent starting point. This knowledge can be tailored to fit the specific needs of each municipality and its residents, and it can also be transferred to other localities in the form of best practices to adapt.

The interviews conducted for this study indicate that urban planners and municipal representatives generally demonstrate a willingness to explore the integration of gamified elements into their public engagement strategies. However, they currently face challenges regarding the practical knowledge and resources necessary to implement these elements effectively.

A significant finding from the interviews indicates that participants predominantly associate gamification with educational environments rather than with urban planning. This limited exposure to gamification within the community engagement framework may contribute to their reluctance to consider it an effective tool for enhancing public participation.

Despite initial skepticism, the interviewees acknowledged the potential advantages of gamification, including enhanced citizen engagement and increased participation in the planning process. Nevertheless, they expressed concerns regarding the lack of knowledge and adaptable best practices for implementing gamification in a manner that is tailored to the specific needs of their municipalities.

The interviews also underscored the significant potential of civil society organizations in supporting local governments with their gamification initiatives. These organizations typically have extensive experience with gamified methodologies and are well positioned to share their knowledge and expertise, thereby aiding in developing tailored solutions for specific municipalities.

The interviews demonstrate an increasing interest in applying gamification to enhance public participation in urban planning. However, the widespread implementation of such strategies is currently impeded by a lack of know-how and resources (both human and financial). Thus, the transfer and adaptation of the existing knowledge, combined with enhanced collaboration between local governments and civil society organizations, can address this deficiency and unlock the full capabilities of gamified approaches in fostering community engagement and promoting urban development.

5. Discussion

This research investigates the status of public participation in urban planning within Hungary, specifically examining the potential role of gamification in enhancing stakeholder engagement. The findings indicate that while there is a general openness among stakeholders to consider the application of gamification in urban planning initiatives, there exists a significant lack of familiarity and guidance regarding its practical implementation. This finding also aligns with the previous research (e.g., [7]).

The advancement of collaborative efforts among participants is considerably impeded by a pervasive atmosphere of mutual distrust, as revealed by the interview findings. This challenge is not solely confined to Hungary; the existing literature indicates that demotivation and distrust are widespread internationally [10]. Nevertheless, the literature

suggests that gamification possesses the potential to create platforms that facilitate closer interactions among stakeholders, thus mitigating misunderstandings [31,32].

The current body of literature underscores the critical necessity for planning processes to be more responsive to emerging challenges and stakeholder inputs [13]. Furthermore, the insights derived from the recent interviews illuminate the importance of clear and effective communication in fostering meaningful participation among all the parties involved. This emphasizes the need for efficient communication channels to ensure that diverse perspectives are accounted for and integrated within the planning framework.

Using gamification to enhance public engagement and participation has gained increasing importance in Western Europe, the US, and Hungary [51]. However, due to the distinctive post-socialist context prevalent in Central European countries [11], the adoption of these progressive strategies remains in its early stages, as evidenced by the interviews conducted.

Research demonstrates that incorporating gamified elements into participatory processes can significantly enhance citizen engagement [3,34,36]. This assertion is corroborated by insights from interviews indicating that participants respond positively to the incentives and enjoyment provided by these elements, ultimately promoting more active and meaningful involvement in civic activities.

The interviewees identified a range of motivational techniques currently employed to stimulate public participation, including idea submissions, competitions, and workshops, which was already explained in the literature [50,51], indicating that applying such techniques cannot be considered a novelty. However, implementing more comprehensive gamified approaches appears to be limited, with only a few examples noted in the capital city of Budapest. Moreover, in many instances, these applications occur without the conscious recognition of their gamified characteristics. This suggests that many practitioners are utilizing gamification strategies to enhance engagement and motivation while remaining unaware of the foundational principles that underpin this methodology.

Our results align with previous findings that highlight the significance of financial resources [77]. Thus, the lack of funds is a motivation and an obstacle to gamifying urban planning processes simultaneously. However, the lack of funds endangers the existing or planned participatory projects, such as the cases of local participatory budgeting initiatives demonstrate.

Additionally, the interviews highlighted the potential role of civil society organizations in assisting local governments with their gamification efforts. These organizations often possess significant experience with gamified approaches and could share their expertise to help formulate tailored solutions for specific municipalities. Notably, the literature does not address the role of civil society organizations, leaving open the question of whether this is a uniquely Hungarian phenomenon or a gap that necessitates further examination in the literature.

The existing literature does not discuss the concept of a gamification handbook or formal guidelines; however, it documents the various challenges individuals may encounter [38]. Insights from interviews suggest that developing such a guide could effectively alleviate these challenges, presenting detailed best practices, the possibilities and challenges of adaptation, and the required human and financial resources.

Furthermore, the cases of gamification implemented in urban planning, as presented in this study, are primarily located in major municipalities [3,34,36], and the interviews were conducted with professionals from these larger cities. This indicates a significant need for gamified approaches, particularly in larger communities, due to the prevalence of social alienation. Our research also supports this statement.

Gamification is undoubtedly an innovative approach; however, like any new method, it has its drawbacks and is not a one-size-fits-all solution for the challenges found in urban environments (Figure 8). While applying gamified elements is significant, careful planning is essential. Without it, gamification can become a burden rather than a benefit.

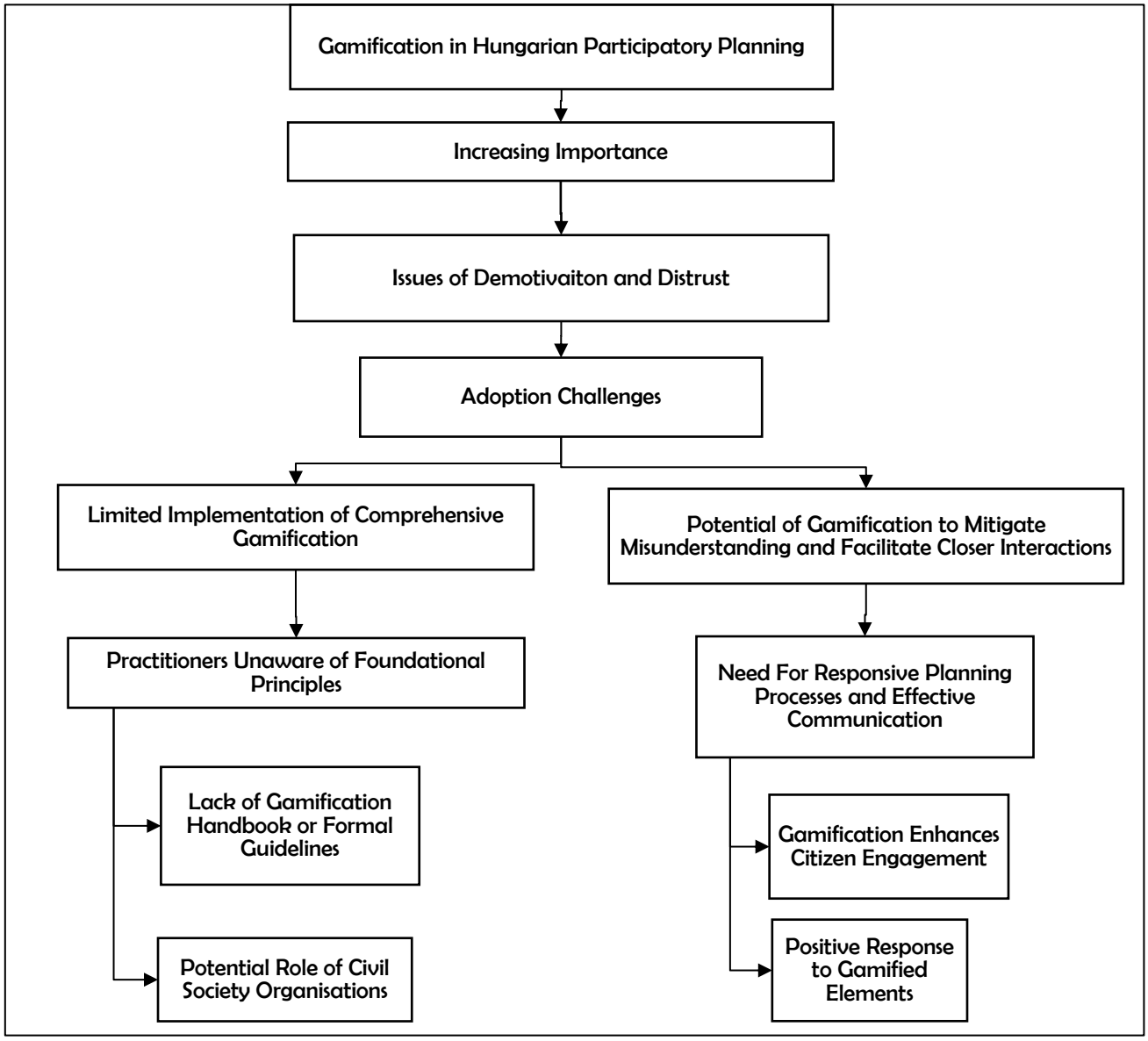


Figure 8. The social environment of gamified urban planning in Hungary. Source: elaborated by the authors.

The study demonstrated that participatory practice in Hungary is in its early stages. However, the experience gained and the obstacles identified can benefit other countries with a low level of participation. Therefore, while Hungary may not be the ideal model of citizen involvement overall, it provides valuable insights into pathfinding and identifying the challenges related to participation and gamification.

6. Conclusions

Our research had three questions to answer. The first focused on the current state of participatory urban planning in Hungary. The results show that participatory approaches are relatively novel in Hungary, and municipalities must deal with several issues when implementing them. Balancing between various interests and finding the appropriate

channel for communication are among the key problems to address. Thus, the development of such approaches is slower than expected or desired. This situation makes the use of gamification quite challenging. With these circumstances, our second question aimed to understand the concern regarding using gamification techniques in Hungarian urban planning. According to our results, gamification is not an unfamiliar concept among decision-makers, and the attitudes toward gamification are positive. Mostly, the larger cities use gamification to enhance participation because social cohesion is usually stronger in smaller communities. The second question focused on the obstacles to using gamification in Hungarian urban planning. According to the interviews, there is a knowledge gap regarding the implementation of gamification. Thus, practical guidelines, toolkits, and exchanging experiences and ideas will be crucial in the future use of gamified planning. Furthermore, the lack of financial and human resources also hinders the use of participatory approaches and gamification techniques.

Since fostering participation is a relatively new and innovative methodology in Hungary, most municipalities are still in the early stages of trying to implement it. Unfortunately, many of them are unsure of how to proceed and struggle with finding the required financial and human resources. This creates a scenario where we must find a path forward, especially as we adopt methods that have been successful in several European (e.g., Germany [7], Sweden [37], Italy [3], Spain, and Finland [35]) countries. In Central Europe, particularly in Hungary, the post-socialist legacy poses unique challenges, such as the low level of civil activity, general distrust towards decision-makers, or hesitancy towards external ideas and solutions. It explains why there is hesitancy regarding gamification, as it is also a new and not widely understood concept. However, several municipalities are already open to new approaches and are attempting to implement participatory methods, including some gamified techniques. But there is still much to learn. Professionals and civil organizations that have successfully organized participatory and gamified events can be crucial in guiding these efforts. Their experience can help enhance methods, foster innovation, and strengthen public and local government relationships. Furthermore, involving civil organizations can also prevent the over-politicization of participatory actions. As the media analysis revealed, some of the critiques towards participatory methods consider these approaches parts of political campaigns. Civic organizations, acting as intermediaries, can distance actions from the political scene.

Our study, on the one hand, contributes to the understanding of the factors influencing gamification in urban planning, shedding light on the significance of post-socialist legacy. Distrust, skepticism, and the apathy of citizens are significant obstacles to urban planning in several post-socialist countries. Since participatory approaches are relatively new planning tools in these countries, municipalities often struggle to find appropriate methods and forms of communication. Enhancing citizen participation through gamification can contribute to more inclusive urban planning and overcome trust issues. By engaging locals, an exchange of knowledge can occur, which can make the processes not just more effective but more transparent as well. Gamification can motivate locals' involvement by expressing their opinions, sharing and generating ideas, or volunteering. This is particularly important when municipalities have minimal resources; thus, enhancing efficiency and providing adequate answers to local needs are complex tasks. Our results show that the lack of financial resources is a motivation and an obstacle for municipal participatory planning processes and gamification. On the other hand, the research provides practical insights regarding the necessity of adaptable know-how to manage the problems related to the lack of knowledge and skill sets. Involving civil organizations can bring expertise and build a bridge between decision-makers and citizens. The citizens' organizations are bottom-up initiatives that embody local aspirations. Thus, they are often more trusted and recognized

by locals. This offers opportunities to overcome the difficulties arising from the erosion of trust between public institutions. Since local governments frequently struggle with financial and human resources, they cannot transfer existing knowledge—external actors, such as civil organizations, can support them to manage this problem. However, the centralization tendencies and politicization of the civil sphere in Hungary may limit the possibilities of such operations. Regarding communication channels, social networks and mobile phone applications are valuable tools. Still, it is essential to highlight that certain social groups can be omitted due to lacking digital skills or not being involved in social networks. Thus, despite applications and social networks seeming to be convenient and cheap means of citizen engagement, face-to-face interactions remain essential. Reaching out on Facebook or homepages is often a one-way communication and does not bring the expected results—it only creates an illusion of citizen dialog and participation. Furthermore, the transfer of know-how through handbooks or guidelines presenting best practices of participatory practices and gamification can contribute to successfully adapting these techniques, helping to overcome knowledge gaps. While compiling and analyzing best practices should be coordinated by national and/or regional actors (e.g., ministries, regional development agencies, etc.), the handbooks should be used mainly at the municipal level, where the best practices and recommendations can be tailored to local needs.

Like any piece of research, this study also has certain limitations. First, the analysis focuses on one country—thus, one should be careful with the generalization of the results. Furthermore, the results are based on interviews and media analysis, which provide valuable insight into the experiences and attitudes of decision-makers but do not provide a comprehensive overview of the topic. Future research should focus on comparing various countries to explore the effects of different social environments on gamification in urban planning. For example, the possibilities regarding the supporting role of civil organizations could be explored in such comparisons. In addition, other methods, such as surveys among residents or content analysis of policy documents, can also be used to gain a more detailed understanding of the issue. Different stakeholder groups can also be interviewed to reveal various points of view on the possibilities of gamification in public participation. Increasing the number of interviewees could incorporate more points of view into the research, thus providing a more nuanced overview of the processes and challenges of participatory urban planning and gamification. The effects of gamified planning techniques (e.g., increased participation and more informed decision-makers and citizens) could also be analyzed—this could provide valuable insights for decision-makers through benchmarking previous actions. Finally, longitudinal studies (e.g., continuing the media analysis and surveying users or decision-makers) are also needed to survey the development of gamification within urban planning.

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